AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 2 · WORKBOOK

A Distribution of Possibilities

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected probability record, with an ordered-representation extension.

Validate, multiply, challenge greedy decoding and normalise with a stated rule.

Keep one habit: Keep prefixes, alternatives, path products, partial event sums and source evidence separate.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Predictive and Generative AI

MAIN LESSON · SECTION 12

W1 · Validate a probability table

> Check alternatives, nonnegative entries and total1.

Your evidence record

- Status first probabilities ready0.5,busy0.3,unknown0.2. End givenready0.8,busy0.5,unknown1; complementary today probabilities0.2,0.5,0. All six two-step paths count toward total1. Separate A/B counterexample first0.6/0.4,end0.51/1. Separate sun/rain first0.55/0.45,end0.4/0.9,complementarymore. Do not mix conditional tables.

- Check 0.5,0.3,0.2 as probabilities. Interpret their sum and the complement of ready.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Calculate Status end paths

> Use each expanded-prefix conditional before adding paths.

Your evidence record

- Status first probabilities ready0.5,busy0.3,unknown0.2. End givenready0.8,busy0.5,unknown1; complementary today probabilities0.2,0.5,0. All six two-step paths count toward total1. Separate A/B counterexample first0.6/0.4,end0.51/1. Separate sun/rain first0.55/0.45,end0.4/0.9,complementarymore. Do not mix conditional tables.

- Compute ready,end; busy,end; unknown,end and their combined probability.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Challenge a greedy claim

> Compare only the specified complete paths.

Your evidence record

- Status first probabilities ready0.5,busy0.3,unknown0.2. End givenready0.8,busy0.5,unknown1; complementary today probabilities0.2,0.5,0. All six two-step paths count toward total1. Separate A/B counterexample first0.6/0.4,end0.51/1. Separate sun/rain first0.55/0.45,end0.4/0.9,complementarymore. Do not mix conditional tables.

- Use A=0.6,B=0.4 and end conditional probabilities 0.51 and 1. Compare greedy A,end with B,end.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Normalise declared nonnegative weights

> Reject a negative or zero-total weight contract rather than silently dividing.

Your evidence record

- Status first probabilities ready0.5,busy0.3,unknown0.2. End givenready0.8,busy0.5,unknown1; complementary today probabilities0.2,0.5,0. All six two-step paths count toward total1. Separate A/B counterexample first0.6/0.4,end0.51/1. Separate sun/rain first0.55/0.45,end0.4/0.9,complementarymore. Do not mix conditional tables.

- Weights are ready 6,busy 3,unknown 1. Convert them into probabilities. What if a weight is -1?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Audit new conditional paths independently

> Keep the first attempt before hints.

Your evidence record

- Use exactly the stated first-token probabilities and end conditionals. Each more probability is1minus its own end conditional. Compare the specified two-step end paths, not all possible eventual completions. The question supplies no real status observation. Record what those missing observations prevent you from claiming.

- First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum.

  Which first token is greedy, which end path is largest, and does either verify a real status?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

My independent explanation, trace and limits

▼ Earlier wording, alternatives and picture transcripts

Independent case: Analyse a new table independently

MAIN LESSON · SECTION 9

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- Explain that Study Buddy responses come from possibilities, so important claims need checks.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Calculate the paths for Next token **distribution**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for Next token **distribution**s. Proposed explanation: “Conditional products assume independence”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A toy next-token **distribution** is sun=.5, rain=.3, snow=.2. Over ten **sample**s, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A toy next-token **distribution** is rain .5, sun .3, snow .2.

Check total and interpret the largest value.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A toy next-token **distribution** assigns cat 0.5, dog 0.3 and bird 0.2. Check the total and name the most likely token. Must it be selected every time when sampling?

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 2 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A toy next-token **distribution** is sun=.5, rain=.3, snow=.2. Over ten **sample**s, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Selected W4 response copy

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum.

Which first token is greedy, which end path is largest, and does either verify a real status?

> **My reasoning and evidence**

▼ Extension · Embeddings and position — W1 Look up vectors

REFERENCE · SECTION 17

Extension · Embeddings and position — W1 Look up vectors

> Use the explicit lookup and position tables beside this task.

Tables, question and response

- Representation extension reference: red embedding[1,0],blue[0,1]; positions0[0,0],1[2,0]. Add equal-width vectors componentwise; preserve ordered rows. Summing rows can lose order. No third position or width-three vector is supplied. IDs are lookup indices, not semantic distances.

- For red blue name the IDs, **embedding** rows and shape before adding positions.

> **My reasoning and evidence**

▼ **Extension · Embeddings and position — W2 Add positions**

REFERENCE · SECTION 18

Extension · Embeddings and position — W2 Add positions

> Use the explicit lookup and position tables beside this task.

Tables, question and response

- Representation extension reference: red embedding[1,0],blue[0,1]; positions0[0,0],1[2,0]. Add equal-width vectors componentwise; preserve ordered rows. Summing rows can lose order. No third position or width-three vector is supplied. IDs are lookup indices, not semantic distances.

- Compute representations for red blue and blue red under the stated position table.

My reasoning and evidence

▼ **Extension · Embeddings and position — W3 Audit the sum**

REFERENCE · SECTION 19

Extension · Embeddings and position — W3 Audit the sum

> Use the explicit lookup and position tables beside this task.

Tables, question and response

- Representation extension reference: red embedding[1,0],blue[0,1]; positions0[0,0],1[2,0]. Add equal-width vectors componentwise; preserve ordered rows. Summing rows can lose order. No third position or width-three vector is supplied. IDs are lookup indices, not semantic distances.

- Sum the rows for both sequences. Does the sum establish that the original text had the same order?

My reasoning and evidence

▼ Extension · Embeddings and position — W4 Trace new vectors independently

REFERENCE · SECTION 20

Extension · Embeddings and position — W4 Trace new vectors independently

> Retained extension task with its own teacher key and response.

Extension question and record

- **Embedding**s A=[1,2], B=[3,0], position 0=[0,1], position 1=[1,0]. Compute represented rows and sums for A B and B A. Explain what the equal sums mean.

> **My reasoning and evidence**

▼ Extension · Embeddings and position — W5 Handle a shape failure

REFERENCE · SECTION 21

Extension · Embeddings and position — W5 Handle a shape failure

> Use the explicit lookup and position tables beside this task.

Tables, question and response

- Representation extension reference: red embedding[1,0],blue[0,1]; positions0[0,0],1[2,0]. Add equal-width vectors componentwise; preserve ordered rows. Summing rows can lose order. No third position or width-three vector is supplied. IDs are lookup indices, not semantic distances.

- An **embedding** has width two but its **position vector** has width three. The **input** also contains a third position absent from the table. Specify honest handling.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

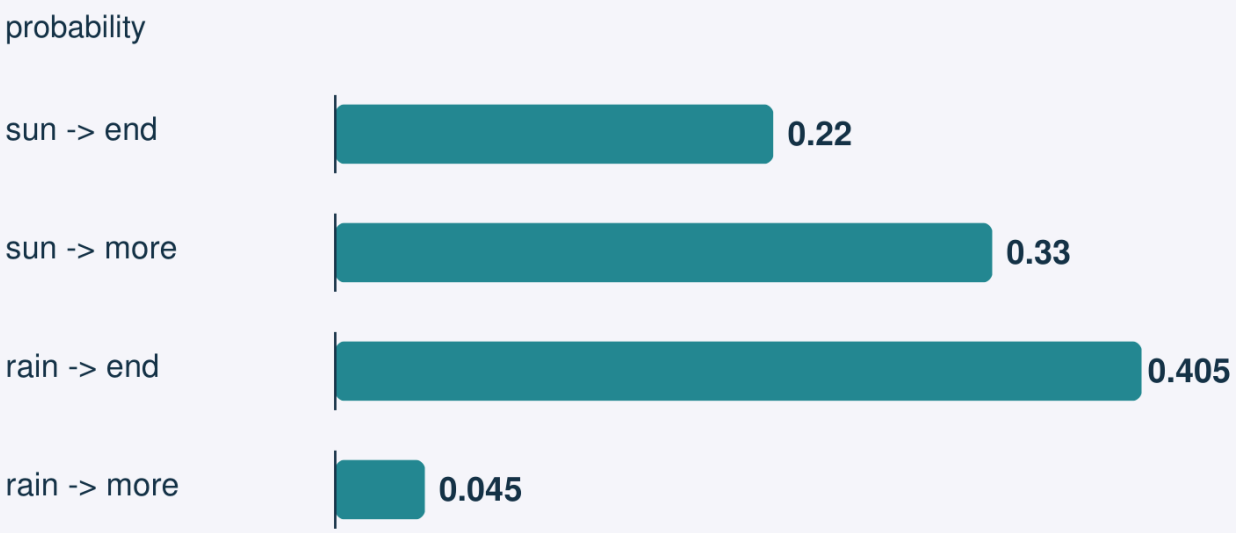
A fictional two-stage token tree illustrates next-token **prediction**.

First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

These are complete toy branches; end paths stop, more paths do not.

AI ACADEMY / AI-3/2

Joint probabilities of the four toy branches



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/2

A Distribution of Possibilities

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First distribution

sun 0.55, rain 0.45: greedy first choice is sun.



2. Sun branches

sun -> end:0.55 x 0.4=0.22; sun -> more:0.55 x 0.6=0.33.



3. Rain branches

rain -> end:0.45 x 0.9=0.405; rain -> more:0.45 x 0.1=0.045.



4. Compare end paths

rain -> end is the more probable of the two complete end paths.

CHANGE ONE THING

Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is 0.625 not a broken probability sum?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Pictures, feedback and repair**

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional two-stage token tree illustrates next-token **prediction**.
- Which labelled step matches this detail? rain→end is the more probable of the two complete end paths.
- 1. Rain branches
- 2. Compare end paths
- 3. Sun branches
- 4. First **distribution**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after trying**

After your attempt, compare with the supported explanation: Compare end paths rain→end is the more probable of the two complete end paths. Conditional probabilities describe what follows a particular prefix.

Multiplying along a path gives its joint probability. The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended.

Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Changed-case question

- Changed-case practice: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. sun→end becomes 0.44, beating unchanged rain→end 0.405.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported explanation: sun→end becomes 0.44, beating unchanged rain→end 0.405.

Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Repair and keep evidence

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

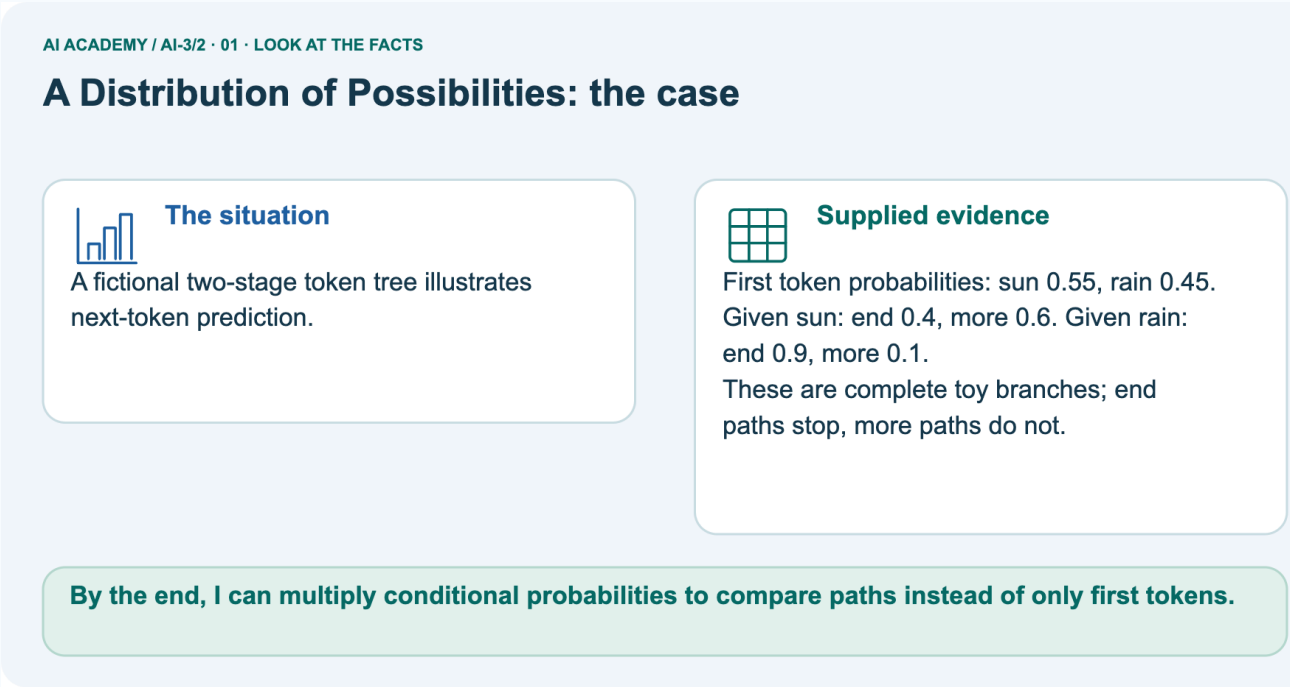
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture rather than repeating every task.

01. A Distribution of Possibilities: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

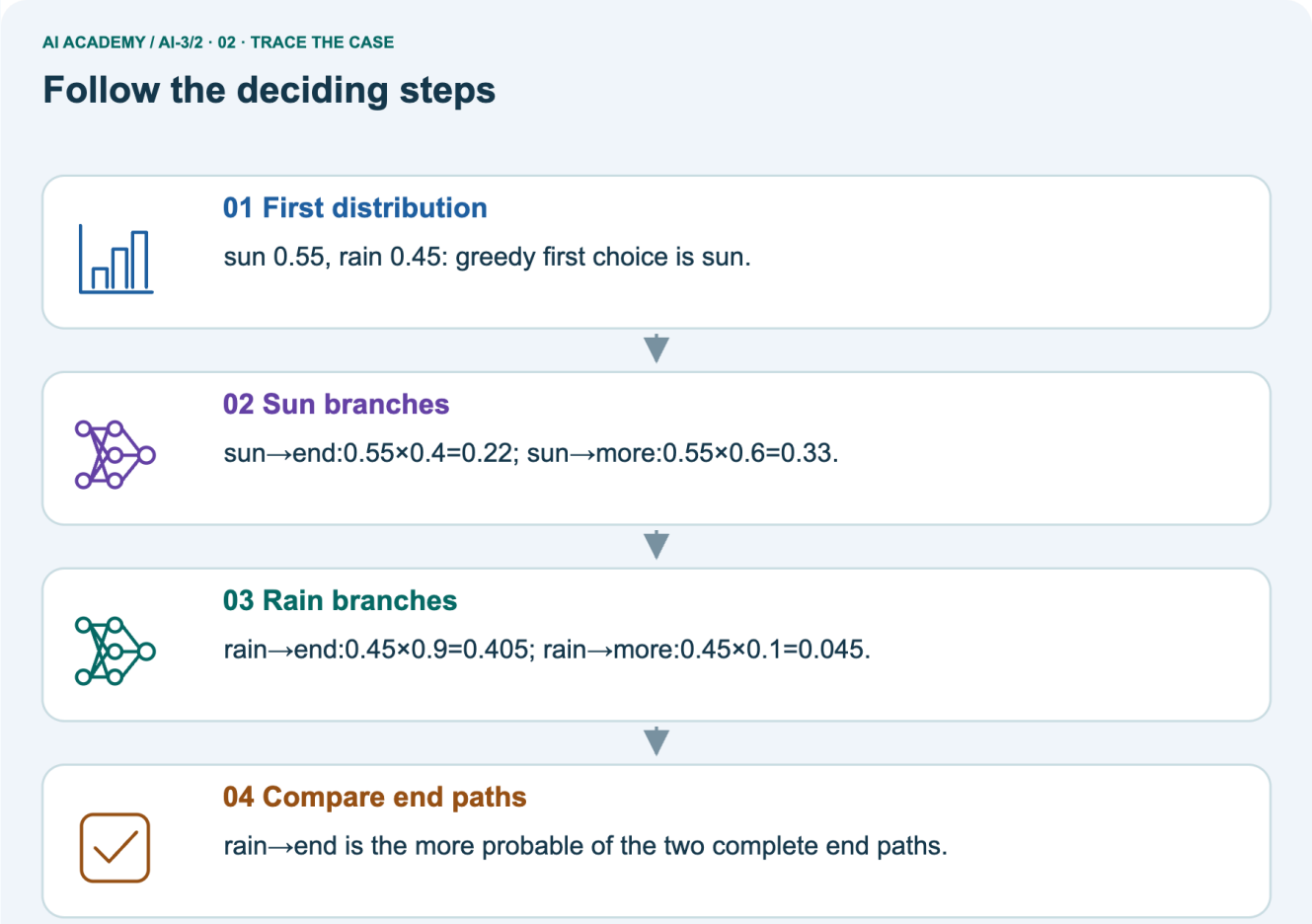
A fictional two-stage token tree illustrates next-token **prediction**. Supplied facts: First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

These are complete toy branches; end paths stop, more paths do not.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



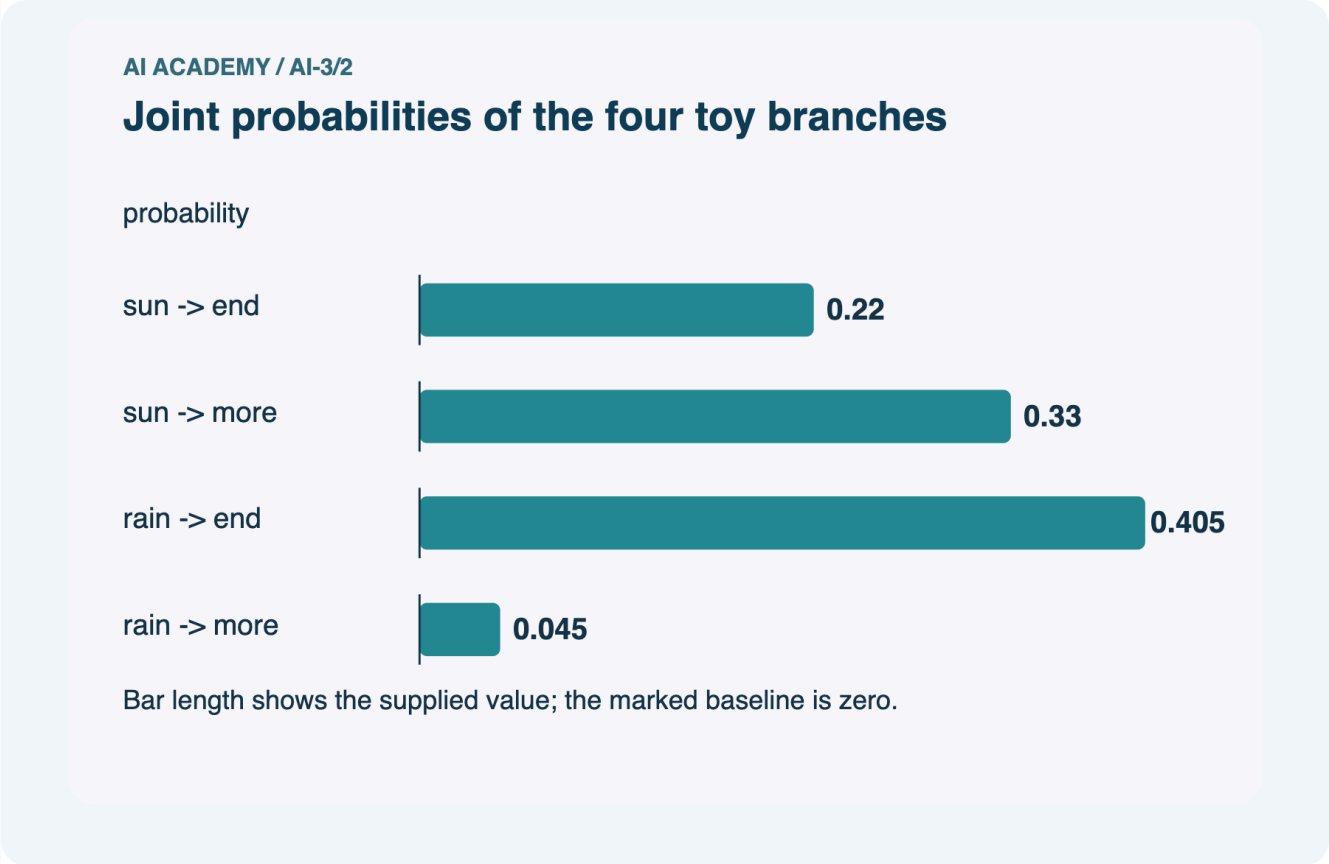
Follow the labelled picture with the matching explanation. Open the vector diagram

First **distribution**: sun 0.55, rain 0.45: greedy first choice is sun. → Sun branches: sun→end:0.55×0.4=0.22; sun→more:0.55×0.6=0.33. → Rain branches: rain→end:0.45×0.9=0.405; rain→more:0.45×0.1=0.045. → Compare end paths: rain→end is the more probable of the two complete end paths.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Joint probabilities of the four toy branches



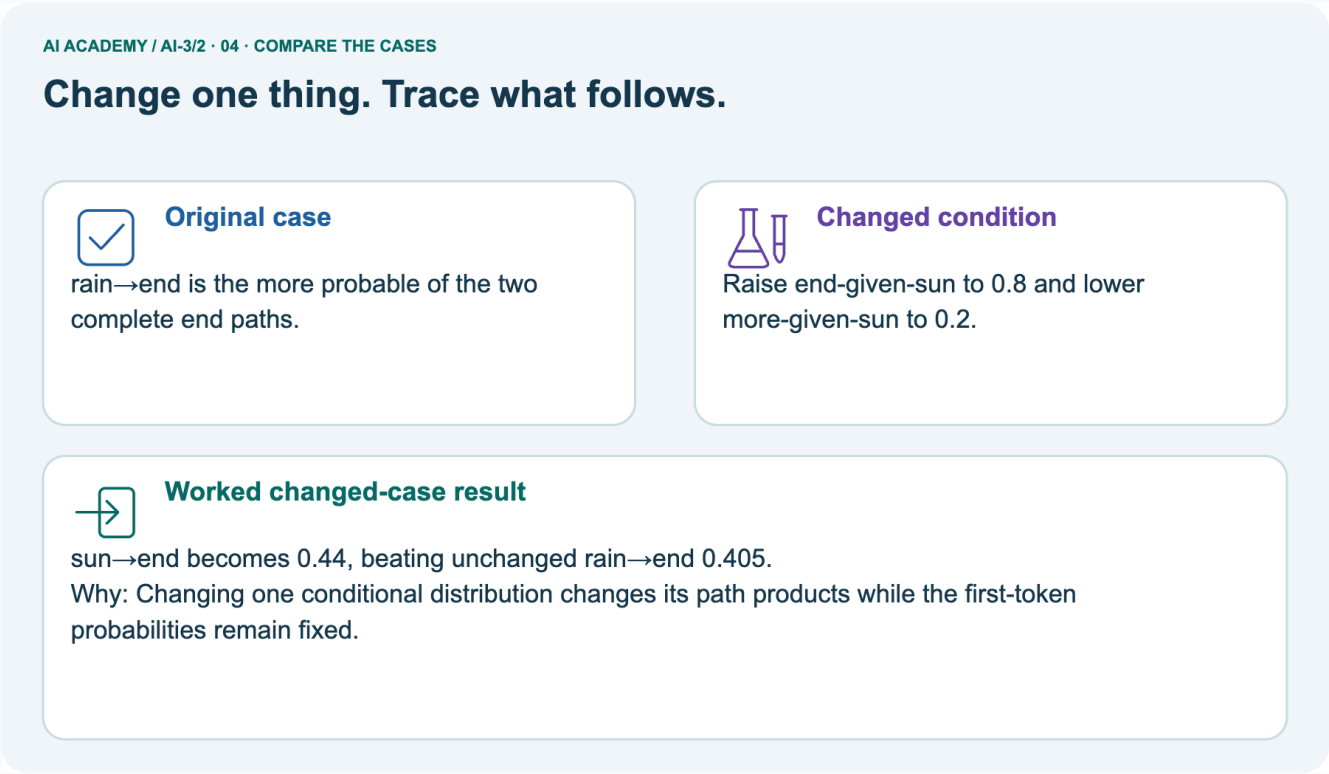
Follow the labelled picture with the matching explanation. Open the vector diagram

Joint probabilities of the four toy branches. Units: probability. sun→end: 0.22. sun→more: 0.33. rain→end: 0.405. rain→more: 0.045.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

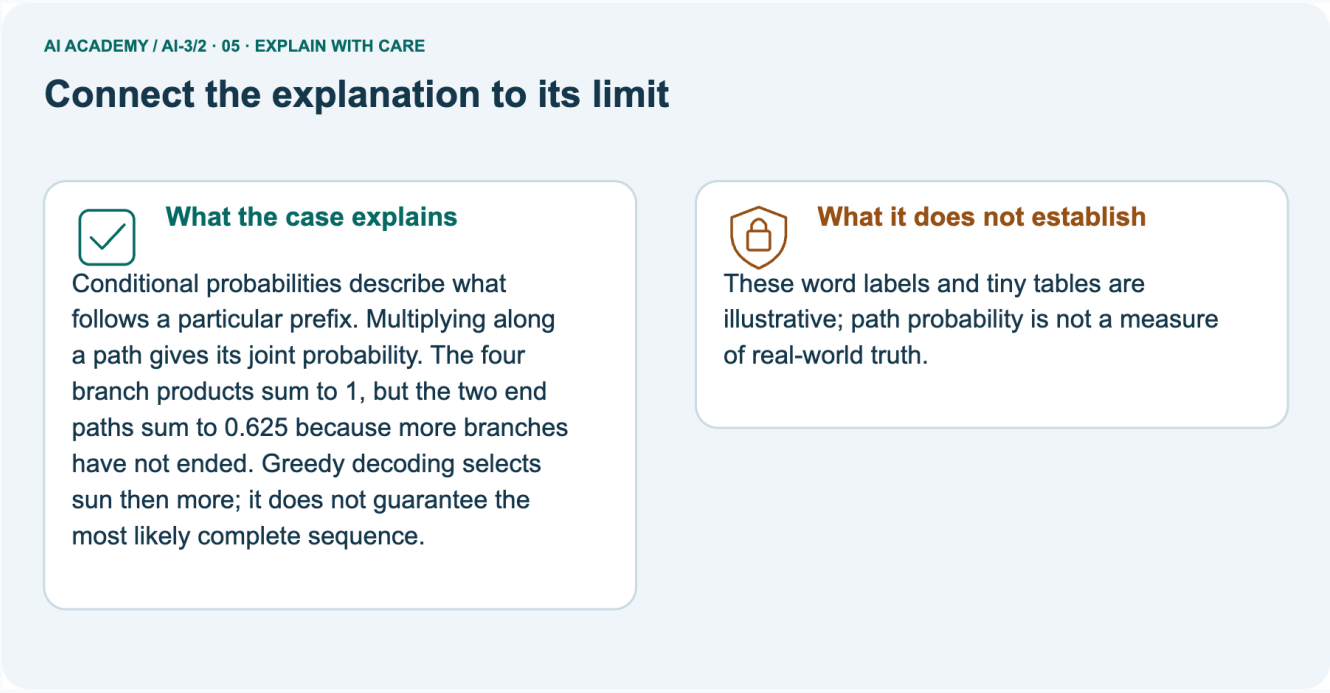
Original case: rain→end is the more probable of the two complete end paths. Changed condition: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Expected result: sun→end becomes 0.44, beating unchanged rain→end 0.405. Reason: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability.

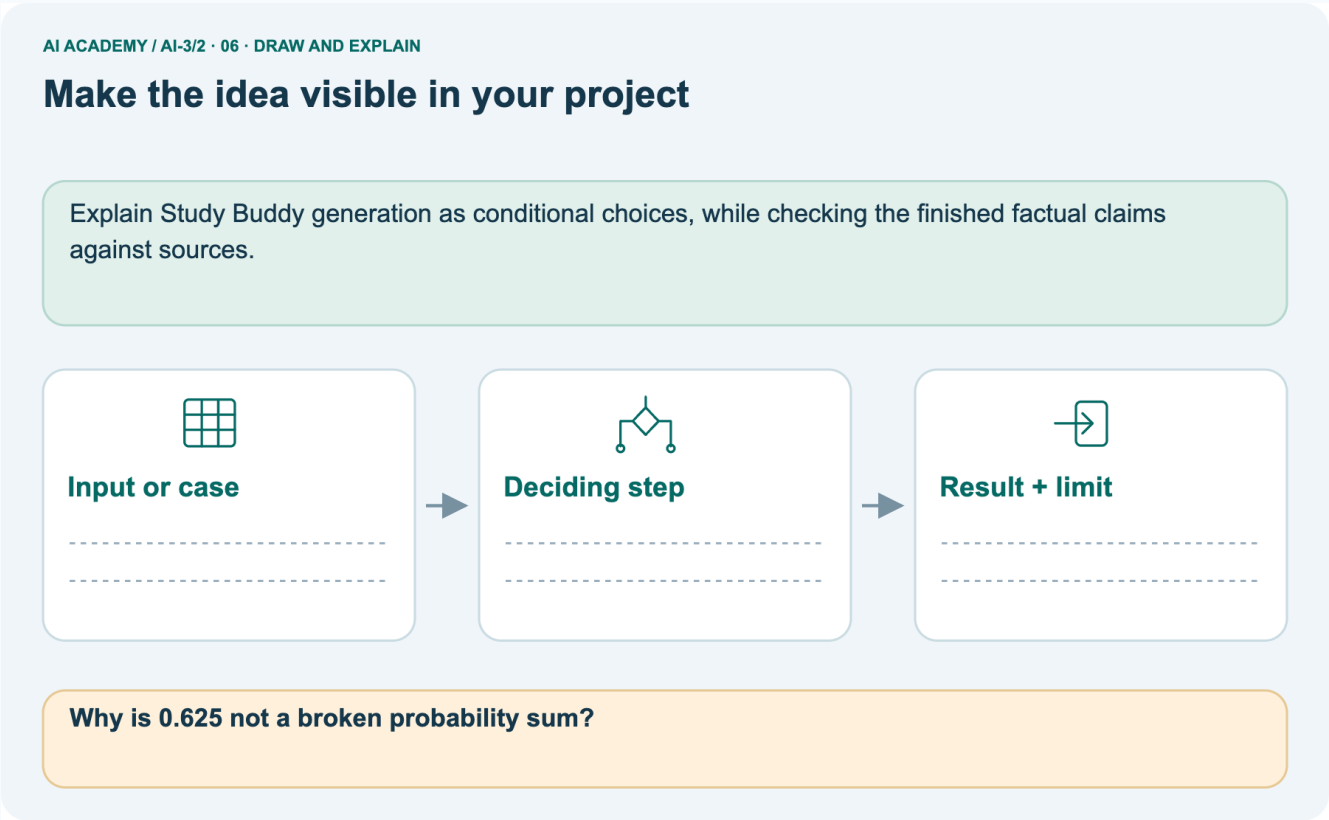
The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Limit: These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources. Draw your **input**, deciding step, result and limit.

Supported question: Why is 0.625 not a broken probability sum?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. A Distribution of Possibilities: the case

02. Follow the deciding steps

03. Joint probabilities of the four toy branches

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.